

Student No.

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Name: _____

Class: 11MTA21_____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2021

YEAR 11

AP2

MATHEMATICS ADVANCED

Time allowed – 1 hour and 30 minutes

**General
Instructions**

- Attempt all questions and clearly label them
- Write your full name on each page of your solutions
- Write using blue or black pen
- NESA approved calculators may be used
- The NESA reference sheet has been provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations

**Total marks:
50**

Section I – 10 marks (pages 2 – 6)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 40 marks (pages 7 – 18)

- Attempt Questions 11-24
- Allow about 1 hour and 15 minutes for this section

Section I

10 marks

Attempt questions 1 - 10

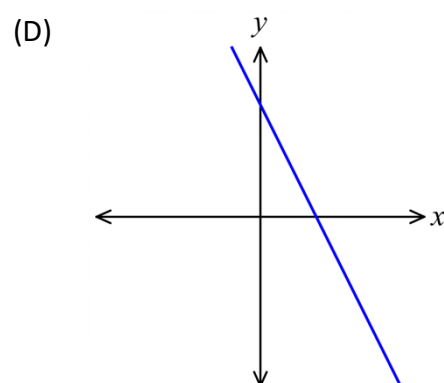
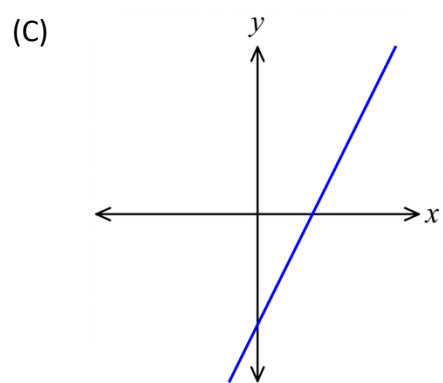
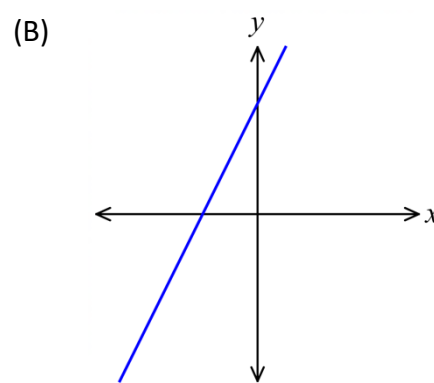
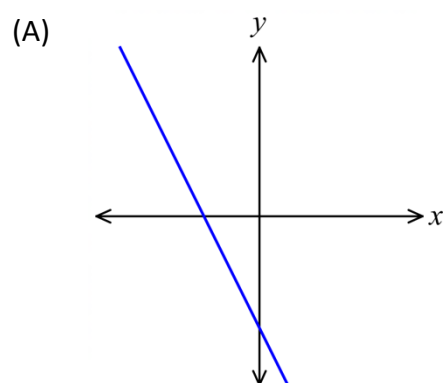
Allow about 15 minutes for this section

Clearly indicate your multiple-choice answers in your solutions.

1. Which of the following is true for the function $f(x) = \frac{3}{x^2} - 2$?

- (A) Even function
- (B) Odd function
- (C) Zero function
- (D) Neither an even, odd or zero function

2. Which one of the following graphs could represent the line with equation $2x - y = 3$?



3. Evaluate $\log_7 \frac{1}{49}$.

- (A) -7
- (B) -2
- (C) 2
- (D) 7

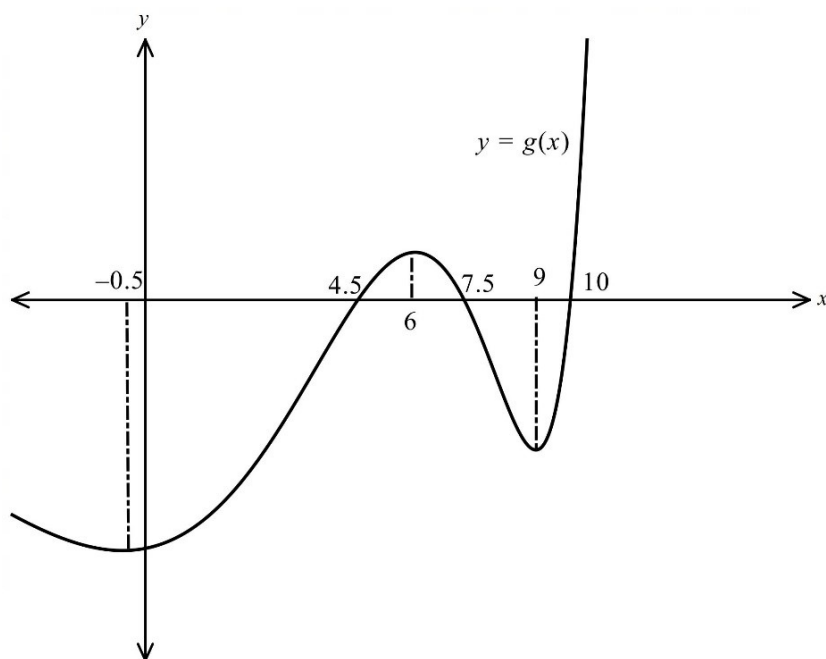
4. Solve the equation $\frac{4}{\sqrt{x}} - 9 = 0$

- (A) $x = \frac{16}{81}$
- (B) $x = \frac{8}{18}$
- (C) $x = \frac{2}{3}$
- (D) $x = 6$

5. What is the exact value of $\sin 135^\circ + \frac{1}{\sin 60^\circ}$?

- (A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$
- (B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$
- (C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$
- (D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

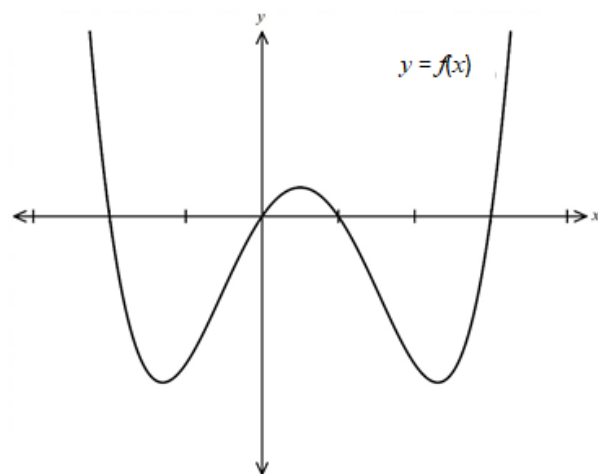
6. The graph of $y = g(x)$ is shown below.
For what values of x is $g(x)$ increasing?



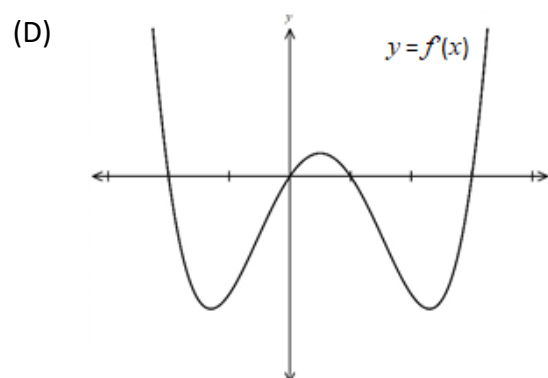
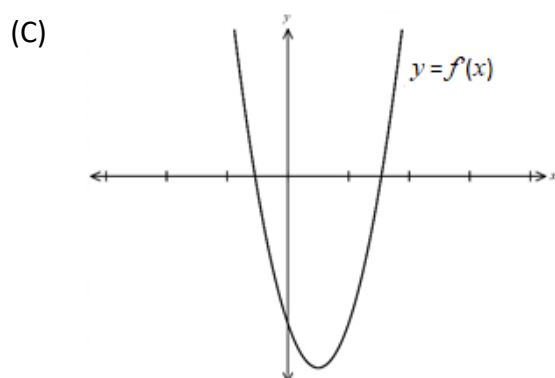
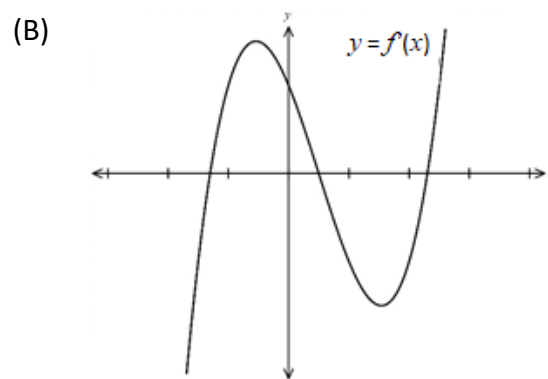
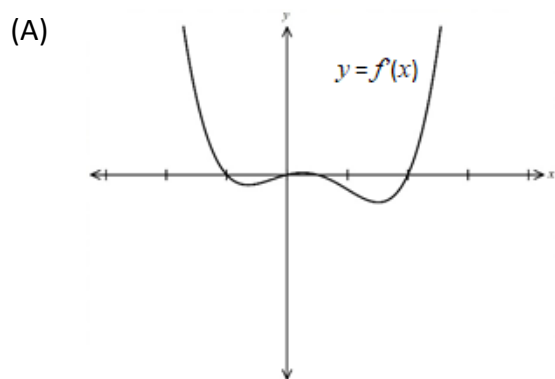
- (A) $(-0.5, 6) \cup (9, \infty)$
- (B) $(-0.5, 4.5) \cup (7.5, \infty)$
- (C) $(4.5, 7) \cup (10, \infty)$
- (D) $(-\infty, -0.5) \cup (6, 9)$
7. Which of the following is the most simplified with a rational denominator equivalent to $\frac{\sqrt{208}}{\sqrt{108}}$?

- (A) $\frac{\sqrt{52}}{\sqrt{27}}$
- (B) $\frac{2\sqrt{39}}{9}$
- (C) $\frac{2\sqrt{13}}{3\sqrt{3}}$
- (D) $\frac{\sqrt{156}}{3}$

8. The graph of $y = f(x)$ is shown below.



Which graph could represent $y = f'(x)$?



9. It is assumed that the number $N(t)$ of ants in a certain nest at time (t) is given by:

$$N(t) = \frac{A}{1+e^{-t}} \text{ where } A \text{ is a constant and } t \text{ is measured in months.}$$

At time $t = 0$, $N(t)$ is estimated at 2×10^5 ants. What is the value of A ?

- (A) 2×10^{-5}
 - (B) 2×10^5
 - (C) 4×10^{-5}
 - (D) 4×10^5
10. The function $f(x) = \ln(2x^{-1})$ can be graphed by applying which transformations to the graph of $g(x) = \ln x$?
- (A) A reflection in the y-axis, then a vertical translation of $\ln 2$
 - (B) A reflection in the x-axis, then a vertical translation of $\ln 2$
 - (C) A reflection in the x-axis, then a horizontal translation of $\ln 2$
 - (D) A reflection in the y-axis, then a horizontal translation of $\ln 2$

End of Section 1

Section II**40 marks****Attempt all questions****Allow about 1 hour and 15 minutes for this section**

- Instructions**
- Answer the questions on blank paper or in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Ensure your questions are clearly labelled and submitted in the correct order.
-

Question 11 (2 marks)**Marks**

Given that $f(x) = 2x^2 - x$ and $g(x) = 2x - 3$, write an expression for $f(g(x))$ given in the form $ax^2 + bx + c$, where a , b and c are integers.

2

Question 12 (2 marks)

Solve the equation

2

$$5\log_{10}x = 3 + 2\log_{10}x$$

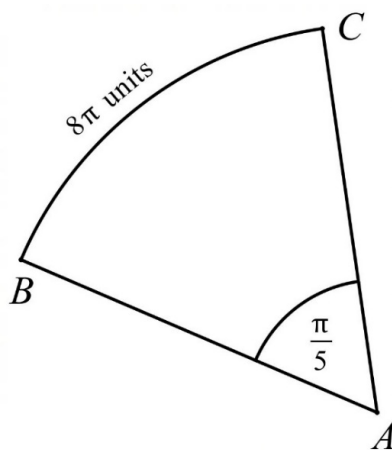
Question 13 (2 marks)

Marks

A sector of a circle is shown below. Point A is the centre of the circle and $\angle BAC = \frac{\pi}{5}$ radians.

2

The arc BC has a length of 8π units.



Find the area of the sector. (Give your answer in terms of π)

See over for Question 14

Question 14 (1 mark)**Marks**

Given $\sin \theta = \frac{7}{12}$ for an obtuse angle θ , find the exact value of $\cos \theta$.

1

Question 15 (2 marks)

Solve $|2x - 3| = 15$

2

See over for Question 16

Question 16 (3 marks)

Marks

Use algebra to solve the following simultaneous equations:

3

$$y = x - 2$$

$$y^2 + x^2 = 10$$

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See over for Question 17

Question 17 (4 marks)

Marks

Let $3x^2 + 12x - 24 = 3(x + a)^2 + b$ where a and b are constants.

- (a) Find the value of a and b .

2

- (b) Hence, determine the **vertex** and the x – **intercepts** for the function $f(x) = 3x^2 + 12x - 24$. Express your answer for the x – intercepts in the form $c \pm d\sqrt{3}$ where c and d are integers.

2

See over for Question 18

Question 18 (4 marks)**Marks**Differentiate, with respect to x .

(a) $(5x^2 - 3)^4$

2

(b) $\frac{3x^2 - 4}{x + 5}$

2

See over for Question 19

Questions 11-18 are worth 20 marks in total
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Question 19 (3 marks)**Marks**

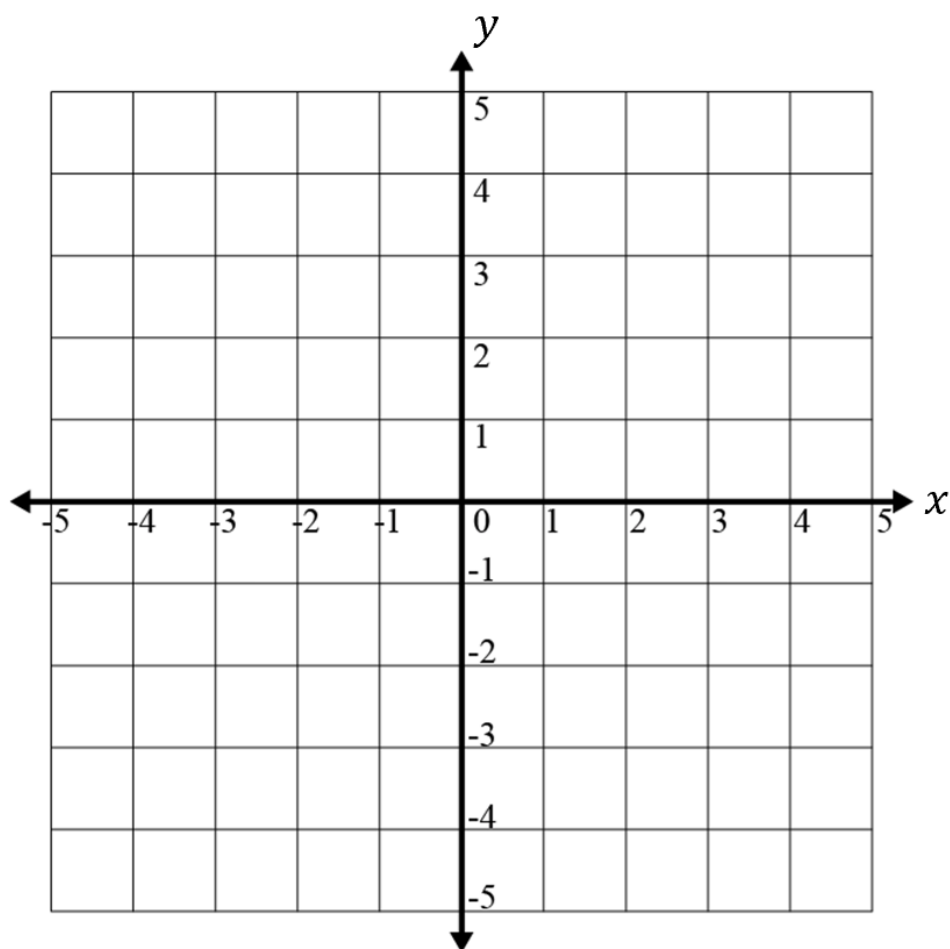
The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} -1 & \text{when } x \leq -2 \\ 2x & \text{when } -2 < x < 0 \\ 3 - x & \text{when } x \geq 0 \end{cases}$$

- (a) Draw a sketch of the graph of $y = f(x)$.

2

Ensure your graph is large enough to clearly label the intercepts and any other relevant values.



- (b) Evaluate $f(-3) + f\left(\frac{1}{2}\right) + 2f(4)$.

1

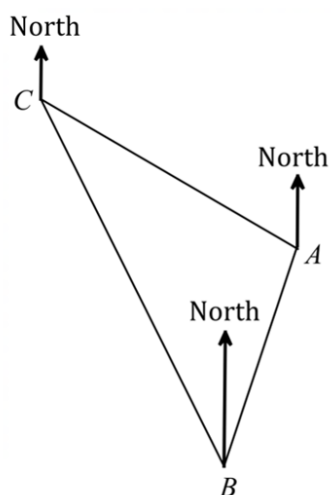
Question 20 (2 marks)**Marks**Differentiate $f(x) = 2x^2 - 3x$ by first principles using**2**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Question 21 (2 marks)Given $\log_a 3 = x$ and $\log_a 5 = y$, find an expression for $\log_a \frac{25}{3}$ in terms of x and y .**2**

Question 22 (4 marks)

Marks



NOT TO SCALE

Lucy walks 4.2km from her house at A on a true bearing of 185°T to the park at B . She then walks on a true bearing of 349°T to the shop at C . Lucy knows she must walk 5.2 km from shop C to her house at A .

- (a) Show that $\angle ABC = 16^\circ$

1

- (b) Hence or otherwise find the bearing of A from C . Answer correct to the nearest degree.

3

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Question 23 (4 marks)**Marks**

A tangent is drawn to the curve $y = x^5 - 3x^2 - 60x$, at the point where $x = -2$.

- (a) Find the gradient of the tangent

2

- (b) Find the equation of the normal to the curve through the same point.

2

See over for Question 24

Question 24 (5 marks)**Marks**

The number of fish, N , in a hatchery at time t months, is given by the formula:

$$N = N_0 e^{0.322t} \text{ where } N_0 \text{ is a constant.}$$

Initially there were 1000 fish in the hatchery.

- (a) Find the number of fish in the hatchery at the end of 8 months.
Answer correct to the nearest hundred.

1

- (b) At the end of which month will the fish population exceed 50 000?

2

Question 24 continues over the page

- (c) Find the rate of increase in the number of fish at the end of six months.
Answer correct to the nearest hundred fish per month.

2

End of Paper

Write your full name on every page of your solutions.

Ensure your questions are in numbered order.

Scan every page of your solutions.

Check that your pdf file includes ALL of your pages.

Upload as one pdf file to Canvas before the submission box closes.

Student No.

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Section I

10 marks

Attempt questions 1 - 10

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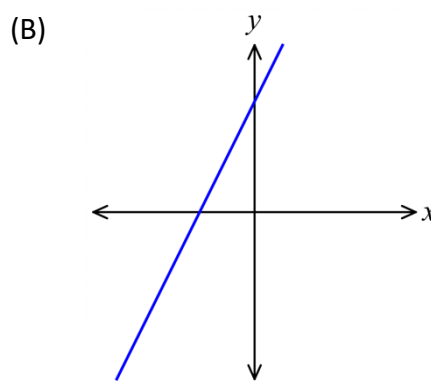
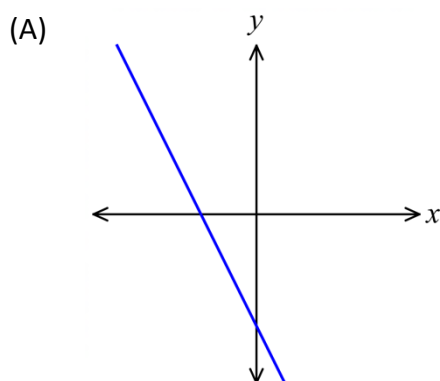
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1. Which of the following is true for the function $f(x) = \frac{3}{x^2} - 2$?

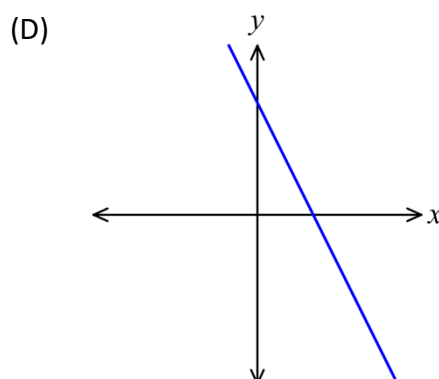
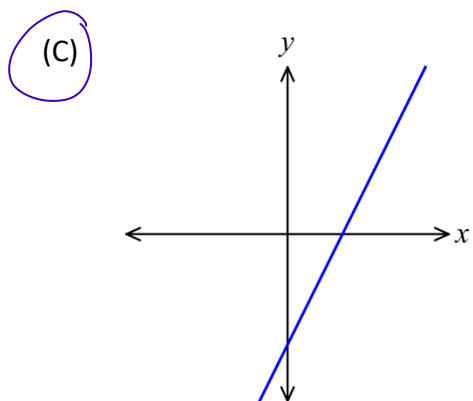
- (A) Even function
(B) Odd function
(C) Zero function
(D) Neither an even, odd or zero function

$$\begin{aligned} f(-x) &= \frac{3}{(-x)^2} - 2 \\ &= \frac{3}{x^2} - 2 \\ &= f(x) \end{aligned}$$

2. Which one of the following graphs could represent the line with equation $2x - y = 3$?



$$2x - 3 = y$$



3. Evaluate $\log_7 \frac{1}{49}$.

- (A) -7
- (B) -2
- (C) 2
- (D) 7

$$\begin{aligned}\log_7 7^{-2} \\&= -2 \log_7 7 \\&= -2\end{aligned}$$

4. Solve the equation $\frac{4}{\sqrt{x}} - 9 = 0$

- (A) $x = \frac{16}{81}$
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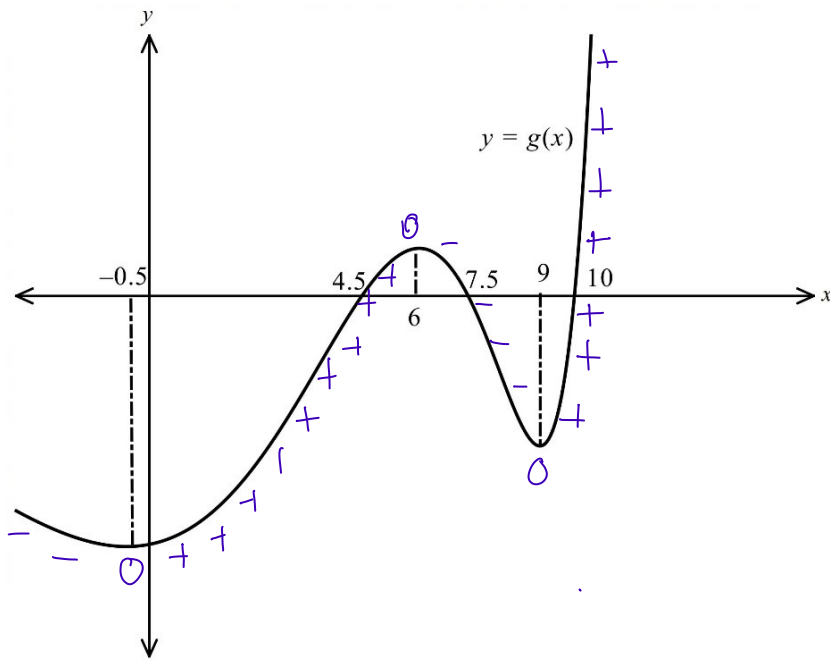
$$\begin{aligned}\frac{4}{\sqrt{x}} &= 9 \\ \sqrt{x} &= \frac{4}{9} \\ x &= \frac{16}{81}\end{aligned}$$

5. What is the exact value of $\sin 135^\circ + \frac{1}{\sin 60^\circ}$?

- (A) $\frac{2\sqrt{2} - \sqrt{3}}{\sqrt{6}}$
- (B) $\frac{2\sqrt{2} - 1}{\sqrt{2}}$
- (C) $\frac{2\sqrt{2} + \sqrt{3}}{\sqrt{6}}$
- (D) $\frac{2\sqrt{2} + 1}{\sqrt{2}}$

$$\begin{aligned}\sin 45^\circ + \frac{1}{\frac{\sqrt{3}}{2}} \\&= \frac{1}{\sqrt{2}} + \frac{2}{\sqrt{3}} \\&= \frac{\sqrt{3} + 2\sqrt{2}}{\sqrt{6}}\end{aligned}$$

6. The graph of $y = g(x)$ is shown below.
For what values of x is $g(x)$ increasing?

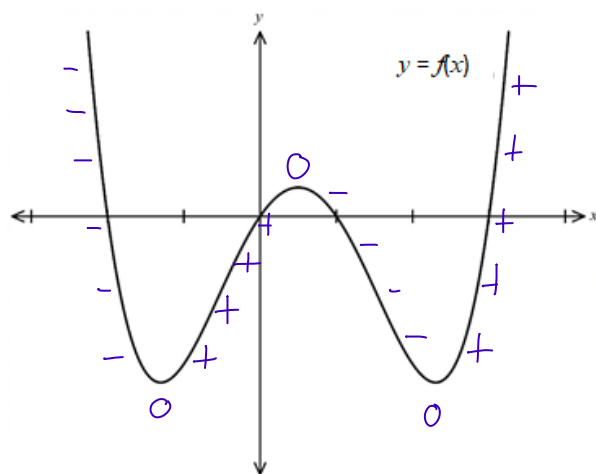


- (A) $(-0.5, 6) \cup (9, \infty)$
- (B) $(-0.5, 4.5) \cup (7.5, \infty)$
- (C) $(4.5, 7) \cup (10, \infty)$
- (D) $(-\infty, -0.5) \cup (6, 9)$
7. Which of the following is the most simplified with a rational denominator equivalent to $\frac{\sqrt{208}}{\sqrt{108}}$?

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- (D) $\frac{\sqrt{156}}{3}$

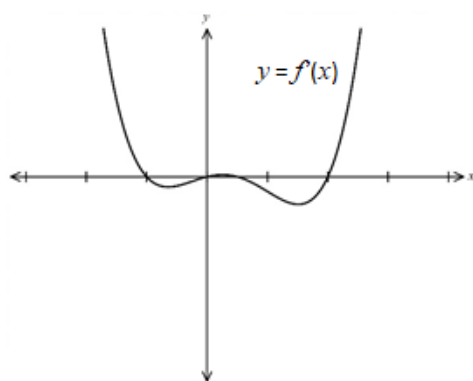
$$\begin{aligned}\sqrt{\frac{208}{108}} &= \sqrt{\frac{52}{27}} \\ &= \frac{2\sqrt{13}}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\ &= \frac{2\sqrt{39}}{9}\end{aligned}$$

8. The graph of $y = f(x)$ is shown below.

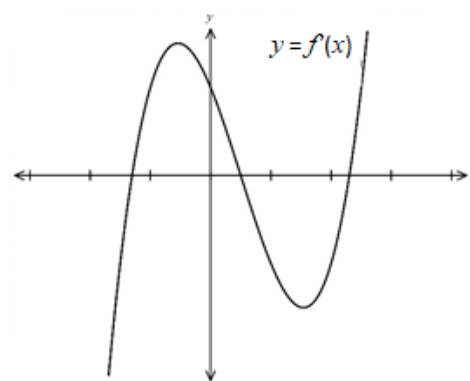


Which graph could represent $y = f'(x)$?

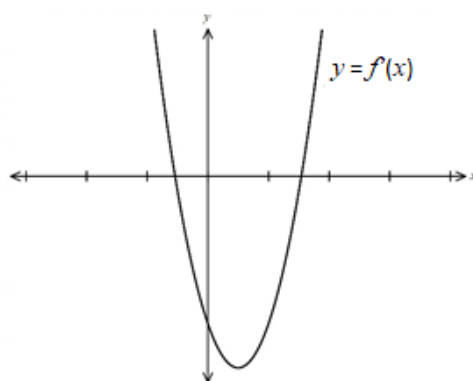
(A)



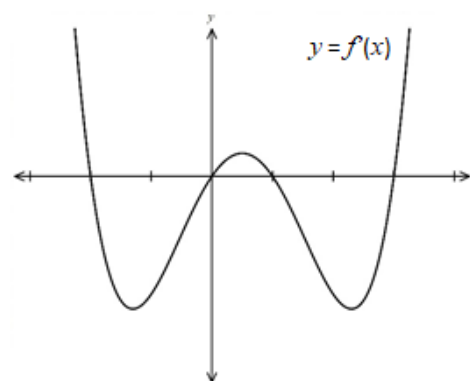
(B)



(C)



(D)



9. It is assumed that the number $N(t)$ of ants in a certain nest at time (t) is given by:

$$N(t) = \frac{A}{1+e^{-t}} \text{ where } A \text{ is a constant and } t \text{ is measured in months.}$$

At time $t = 0$, $N(t)$ is estimated at 2×10^5 ants. What is the value of A ?

- (A) 2×10^{-5}
(B) 2×10^5
(C) 4×10^{-5}
(D) 4×10^5

$$\begin{aligned} \frac{A}{1+e^{-0}} &= 2 \times 10^5 \\ A &= (2 \times 10^5) \times 2 \\ A &= 4 \times 10^5 \end{aligned}$$

10. The function $f(x) = \ln(2x^{-1})$ can be graphed by applying which transformations to the graph of $g(x) = \ln x$?

- (A) A reflection in the y -axis, then a vertical translation of $\ln 2$
(B) A reflection in the x -axis, then a vertical translation of $\ln 2$
(C) A reflection in the x -axis, then a horizontal translation of $\ln 2$
(D) A reflection in the y -axis, then a horizontal translation of $\ln 2$

End of Section 1

$$\begin{aligned} \ln 2x^{-1} &= \ln 2 + \ln x^{-1} \\ &= \ln 2 - \ln x \\ &= -\ln x + \ln 2 \end{aligned}$$

Reflection in the x -axis Translation up $\ln 2$ units.

Section II

40 marks

Attempt all questions

Allow about 1 hour and 15 minutes for this section

- Instructions**
- Answer the questions on blank paper or in the spaces provided. These spaces provide guidance for the expected length of response.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Ensure your questions are clearly labelled and submitted in the correct order.
-

Question 11 (2 marks)

Marks

Given that $f(x) = 2x^2 - x$ and $g(x) = 2x - 3$, write an expression for $f(g(x))$ given in the form $ax^2 + bx + c$, where a , b and c are integers.

2

$$\begin{aligned} f(2x-3) &= 2(2x-3)^2 - (2x-3) \\ &= (2x-3)[2(2x-3) - 1] \\ &= (2x-3)(4x-7) \end{aligned}$$

$$\begin{aligned} \text{OR } f(2x-3) &= 2(4x^2 - 12x + 9) - 2x + 3 \\ &= 8x^2 - 24x + 18 - 2x + 3 \\ &= 8x^2 - 26x + 21 \end{aligned}$$

Question 12 (2 marks)

Solve the equation

2

$$5\log_{10}x = 3 + 2\log_{10}x$$

$$\log_{10}x^5 - \log_{10}x^2 = 3$$

$$\log_{10}\left(\frac{x^5}{x^2}\right) = 3$$

$$\log_{10}x^3 = 3$$

$$x^3 = 10^3$$

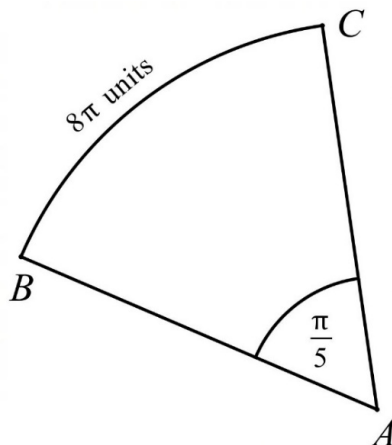
$$\therefore x = 10$$

Question 13 (2 marks)**Marks**

A sector of a circle is shown below. Point A is the centre of the circle and $\angle BAC = \frac{\pi}{5}$ radians.

2

The arc BC has a length of 8π units.



Find the area of the sector. (Give your answer in terms of π)

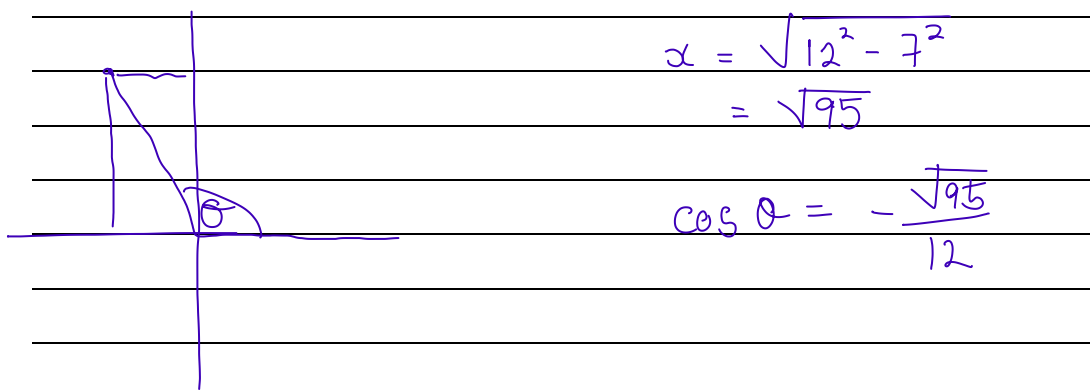
$$\begin{aligned} r \times \frac{\pi}{5} &= 8\pi \\ r &= 8\pi \times \frac{5}{\pi} \\ &= 40 \text{ units} \end{aligned}$$

$$\begin{aligned} \text{Area} &= \frac{1}{2} (40)^2 \left(\frac{\pi}{5} \right) \\ &= 160\pi \text{ units}^2 \end{aligned}$$

See over for Question 14

Question 14 (1 mark)**Marks**

Given $\sin \theta = \frac{7}{12}$ for an obtuse angle θ , find the exact value of $\cos \theta$.

1**Question 15** (2 marks)

Solve $|2x - 3| = 15$

2

$$\begin{array}{lcl} 2x - 3 = 15 & \text{or} & -(2x - 3) = 15 \\ 2x = 18 & & -2x + 3 = 15 \\ x = 9 & & -2x = 12 \\ & & x = -6 \end{array}$$

See over for Question 16

Question 16 (3 marks)**Marks**

Use algebra to solve the following simultaneous equations:

3

$$y = x - 2 \quad (1)$$

$$y^2 + x^2 = 10 \quad (2)$$

Subst (1) in (2)

$$(x-2)^2 + x^2 = 10$$

$$x^2 - 4x + 4 + x^2 = 10$$

$$2x^2 - 4x - 6 = 0$$

$$2(x^2 - 2x - 3) = 0$$

$$2(x-3)(x+1) = 0$$

$$x = 3 \quad \text{or} \quad x = -1$$

(3)

Subst (3) in (1)

$$y = 3 - 2$$

$$y = 1$$

$$(3, 1)$$

or

$$y = -1 - 2$$

$$y = -3$$

$$(-1, -3)$$

See over for Question 17

Question 17 (4 marks)**Marks**

Let $3x^2 + 12x - 24 = 3(x + a)^2 + b$ where a and b are constants.

- (a) Find the value of a and b .

2

$$\begin{aligned} & 3(x^2 + 4x - 8) \\ &= 3[(x^2 + 4x + 4) - 8 - 4] \\ &= 3[(x+2)^2 - 12] \\ &= 3(x+2)^2 - 36 \\ \therefore a &= 2 \quad \text{and} \quad b = -36 \end{aligned}$$

- (b) Hence, determine the **vertex** and the **x – intercepts** for the function $f(x) = 3x^2 + 12x - 24$. Express your answer for the x – intercepts in the form $c \pm d\sqrt{3}$ where c and d are integers.

2

$$\begin{aligned} & \text{Vertex } (-2, -36) \\ & x\text{-int: } 3x^2 + 12x - 24 = 0 \\ & 3(x+2)^2 - 36 = 0 \\ & 3(x+2)^2 = 36 \\ & (x+2)^2 = 12 \\ & x+2 = \pm\sqrt{12} \\ & x = -2 \pm \sqrt{12} \\ & = -2 \pm 2\sqrt{3} \end{aligned}$$

See over for Question 18

Question 18 (4 marks)**Marks**Differentiate, with respect to x .

(a) $(5x^2 - 3)^4$ $\frac{d}{dx} (5x^2 - 3)^4$ 2

$$\begin{aligned} &= 4(5x^2 - 3)^3 \times 10x \\ &= 40x(5x^2 - 3)^3 \end{aligned}$$

(b) $\frac{3x^2 - 4}{x + 5}$ 2

$$\begin{aligned} \text{Let } u &= 3x^2 - 4 & \text{and } v &= x + 5 \\ u' &= 6x & v' &= 1 \end{aligned}$$

$$\begin{aligned} \frac{d}{dx} \left(\frac{3x^2 - 4}{x + 5} \right) &= \frac{6x(x + 5) - 1(3x^2 - 4)}{(x + 5)^2} \\ &= \frac{6x^2 + 30x - 3x^2 + 4}{(x + 5)^2} \\ &= \frac{3x^2 + 30x + 4}{(x + 5)^2} \end{aligned}$$

See over for Question 19

Question 19 (3 marks)**Marks**

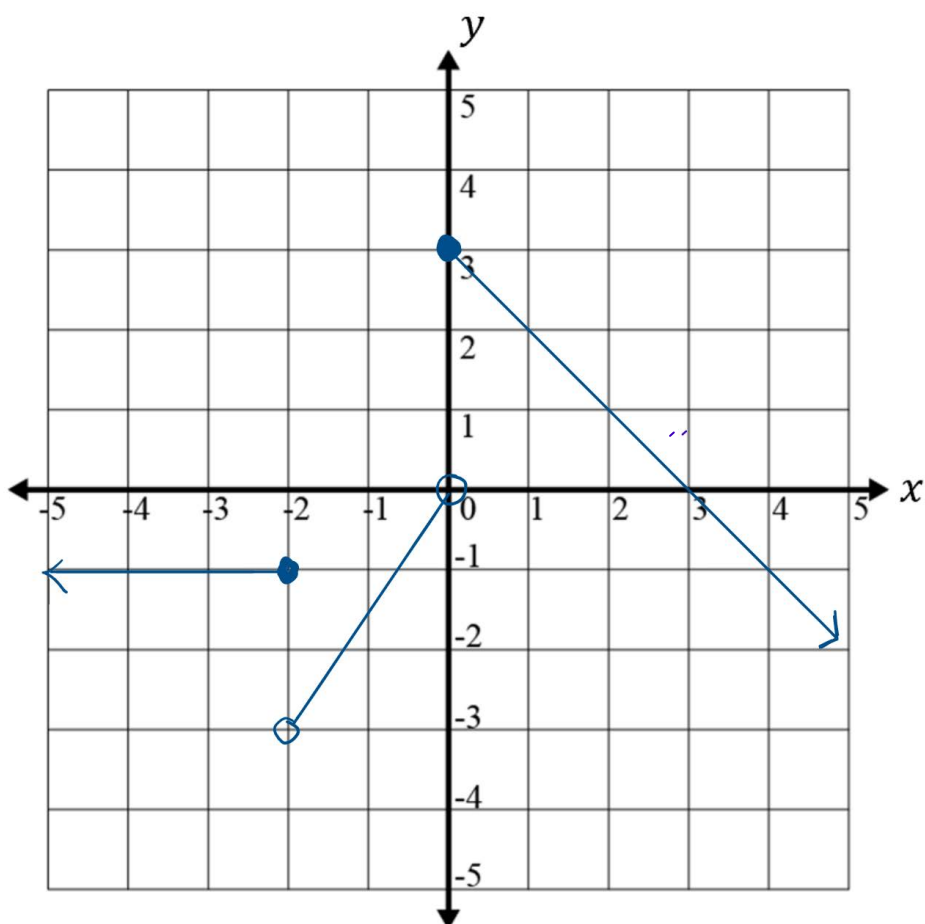
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$$f(x) = \begin{cases} -1 & \text{when } x \leq -2 \\ 2x & \text{when } -2 < x < 0 \\ 3 - x & \text{when } x \geq 0 \end{cases}$$

- (a) Draw a sketch of the graph of $y = f(x)$.

2

Ensure your graph is large enough to clearly label the intercepts and any other relevant values.



- (b) Evaluate $f(-3) + f\left(\frac{1}{2}\right) + 2f(4)$.

1

$$= -1 + 2.5 + 2(-1)$$

$$= 1.5 - 2$$

$$= -0.5$$

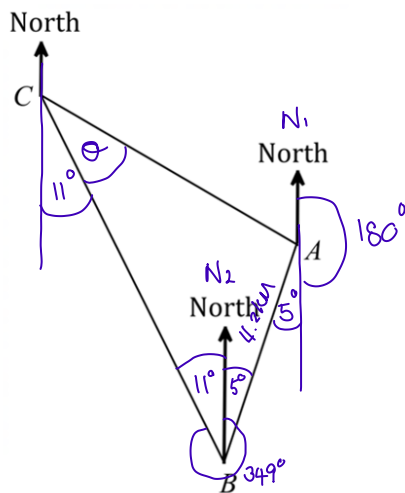
Question 20 (2 marks)**Marks**Differentiate $f(x) = 2x^2 - 3x$ by first principles using**2**

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{2(x+h)^2 - 3(x+h) - (2x^2 - 3x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{2x^2} + 4xh + 2h^2 - \cancel{3x} - 3h - \cancel{2x^2} + \cancel{3x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - 3h}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(4x + 2h - 3)}{\cancel{h}} \\ &= \lim_{h \rightarrow 0} 4x + 2h - 3 \\ &= 4x - 3 \end{aligned}$$

Question 21 (2 marks)**2**Given $\log_a 3 = x$ and $\log_a 5 = y$, find an expression for $\log_a \frac{25}{3}$ in terms of x and y .

$$\begin{aligned} \log_a \frac{25}{3} &= \log_a 25 - \log_a 3 \\ &= \log_a 5^2 - x \\ &= 2 \log_a 5 - x \\ &= 2y - x \end{aligned}$$



Lucy walks 4.2km from her house at A on a true bearing of 185° to the park at B . She then walks on a true bearing of 349° to the shop at C . Lucy knows she must walk 5.2 km from shop C to her house at A .

- (a) Show that $\angle ABC = 16^\circ$ 1

$$\angle N_1BA = 185^\circ - 180^\circ$$

$$= 5^\circ$$

$$\angle N_2BC = 360^\circ - 349^\circ$$

$$= 11^\circ$$

$$\therefore \angle ABC = 11^\circ + 5^\circ$$

$$= 16^\circ$$

- (b) Hence or otherwise find the bearing of A from C . Answer correct to the nearest degree. 3

$$\frac{\sin \theta}{4.2} = \frac{\sin 16^\circ}{5.2}$$

$$\sin \theta = \frac{4.2 \sin 16^\circ}{5.2}$$

$$\theta = \sin^{-1} \left(\frac{4.2 \sin 16^\circ}{5.2} \right)$$

$$= 12.86\dots$$

$$\text{Bearing from } C \text{ to } A = 180 - (11 + 12.86\dots)$$

$$\approx 156^\circ \text{T}$$

Question 23 (4 marks)**Marks**

A tangent is drawn to the curve $y = x^5 - 3x^2 - 60x$, at the point where $x = -2$.

- (a) Find the gradient of the tangent

2

$$\begin{aligned}\frac{dy}{dx} &= 5x^4 - 6x - 60 \\ \text{When } x = -2 \quad \frac{dy}{dx} &= 5(-2)^4 - 6(-2) - 60 \\ &= 80 + 12 - 60 \\ &= 32 \\ \therefore m &= 32\end{aligned}$$

- (b) Find the equation of the normal to the curve through the same point.

2

$$\begin{aligned}m_{\perp} &= -\frac{1}{32} \quad \text{When } x = -2 \\ y &= (-2)^5 - 3(-2)^2 - 60(-2) \\ &= 76 \\ &(-2, 76) \\ y - 76 &= -\frac{1}{32}(x + 2) \\ 32y - 2432 &= -x - 2 \\ x + 32y - 2430 &= 0\end{aligned}$$

See over for Question 24

Question 24 (5 marks)**Marks**

The number of fish, N , in a hatchery at time t months, is given by the formula:

$$N = N_0 e^{0.322t} \text{ where } N_0 \text{ is a constant.}$$

Initially there were 1000 fish in the hatchery.

- (a) Find the number of fish in the hatchery at the end of 8 months.
Answer correct to the nearest hundred.

1

$$\text{When } t = 0 \quad N = 1000$$

$$\therefore N_0 e^0 = 1000$$

$$\therefore N_0 = 1000$$

$$\text{At } t=8 \quad N = 1000 e^{0.322(8)}$$

$$\approx 13100 \text{ fish}$$

- (b) At the end of which month will the fish population exceed 50 000?

2

$$1000 e^{0.322t} = 50\,000$$

$$e^{0.322t} = 50$$

$$0.322t = \ln 50$$

$$t = \frac{\ln 50}{0.322}$$

$$t = 12.149... \text{ months}$$

$$\therefore \text{At the end of month } 13$$

Question 24 continues over the page

- (c) Find the rate of increase in the number of fish at the end of six months.
Answer correct to the nearest hundred fish per month.

2

$$\begin{aligned}
 N &= 1000 e^{0.322t} \\
 \frac{dN}{dt} &= 0.322 \times 1000 e^{0.322t} \\
 &= 322 e^{0.322t} \\
 \text{At } t = 6 \quad \frac{dN}{dt} &= 322 e^{0.322(6)} \\
 &= 2200 \text{ fish/month}
 \end{aligned}$$

End of Paper

Write your full name on every page of your solutions.

Ensure your questions are in numbered order.

Scan every page of your solutions.

Check that your pdf file includes ALL of your pages.

Upload as one pdf file to Canvas before the submission box closes.